COSC 6397 | Topics Computer Science

Section: 26162 | Face-to-Face

Instructor Information

Prof. Gopal Pandurangan
Department: Computer Science
Email: gpandura@cougarnet.uh.edu
Phone: 713/743-5556

Office Hours
By appointment

General Course Information

COSC 6397: **Quantum Computing**

Course Meeting Time: TuTh 10:00AM - 11:30AM

Course Meeting Location: SW 219

Course Objectives and Student Learning Outcomes

Course Objectives:

This course will provide a rigorous, principled introduction to quantum computing, along with associated algorithms and techniques.

The course will teach the fundamentals of quantum computing and present key computer science applications. The course assumes no prior knowledge of quantum mechanics or physics. The course will cover all required concepts. Unlike typical quantum computing courses and textbooks, this course will be taught from a computer scientist's perspective, particularly for those with a background in algorithms. The course will teach quantum computing from a classical randomized algorithmic perspective.

What the students will learn:

By the end of the course, students are expected to gain a solid understanding of the fundamentals of quantum computation.

They will learn important quantum algorithms and their associated correctness proofs. They will also learn important quantum computing phenomena and their applications.

Required Instructional Materials

Recommended references:

1. John Watrous. Lecture Notes in Quantum Computation, University of Calgary, 2006. https://cs.uwaterloo.ca/~watrous/QC-notes/
2. G. Pandurangan. Randomization and Probabilistic Techniques in Computer Science. https://sites.google.com/site/gopalpandurangan/home/randalgos

Course Schedule, Assignments, and Assessments

Lecture Scribes: 25%

Midterm exam 1: 25%

Midterm exam 2: 25%

Final project: 25%

More details on the the above will be discussed in class.

Discussion and Lecture Topics

1. Fundamental mathematical concepts: Probability theory, linear algebra.
2. Fundamentals of randomized algorithms: Sampling, Markov chains, random walks
3. Quantum information: Single and multiple qubits, unitary operations, Dirac notation, tensor product, no-cloning theorem.
4. Quantum Circuits: Superdense coding and quantum teleportation.
5. Deutsch's algorithm, phase kickback.
6. Deutsch-Jozsa algorithm.
7. Simon's algorithm.
8. Grover's algorithm.
9. Phase estimation and Shor's algorithm.
10. Quantum cryptography --- quantum key distribution.
11. Nonlocality and entanglement.
12. Introduction to quantum complexity theory.
13. Introduction to distributed quantum computing.

Grading Rubrics and Weights

Work must be submitted on time. There is a penalty of 2% per hour for late submissions. A submission more than 50 hours late will receive a zero.

The grades are scaled so that the average class score corresponds approximately to a B+ grade.

Requests for review or reconsideration of answers for a regrade must be submitted within one week of the return of the graded work.

Course Policies and Procedures

All submitted work should be your own. Copying or using other people's work (including from the Web or AI) will result in –MAX points, where MAX is the maximum possible number of points for that assignment/homework/exam.

Repeat offenses will result in a failing grade for the course and will be reported to the Chair. If you have any questions, please reach out to the professor.

By submitting your work or exam, you affirm that you have followed the Academic Honesty Policy and the Honor Code.

The materials provided by the instructor in this course are for the use of the students enrolled in the course only. Copyrighted course materials (including lecture notes, homeworks, assignments, exams, and their solutions) should not be further disseminated. This includes sharing content with commercial course material suppliers such as Course Hero or Chegg. Students are also prohibited from sharing materials derived from the instructor's content (e.g., a student's lecture notes).

University Policies and Student Support Resources

Mental Health and Wellness Resources

The University of Houston has a number of resources to support students' mental health and overall wellness, including CoogsCARE and the UH Go App. UH Counseling and Psychological Services (CAPS) offers 24/7 mental health support for all students, addressing various concerns like stress, college adjustment and sadness. CAPS provides individual and couples counseling, group therapy, workshops and connections to other support services on and off- campus. For assistance visit uh.edu/caps, call 713-743-5454, or visit a Let's Talk location in-person or virtually. Let's Talk are daily, informal confidential consultations with CAPS therapists where no appointment or paperwork is needed.

Need Support Now? If you or someone you know is struggling or in crisis, help is available. Call CAPS crisis support 24/7 at 713-743-5454, or the National Suicide and Crisis Lifeline: call or text 988, or chat 988lifeline.org.

Title IX/Sexual Misconduct

Per the UHS Sexual Misconduct Policy, your instructor is a "responsible employee" for reporting purposes under Title IX regulations and state law and must report incidents of sexual misconduct (sexual harassment, non-consensual sexual contact, sexual assault, sexual exploitation, sexual intimidation, intimate partner violence, or stalking) about which they become aware to the Title IX office (known at UH as the Equal Opportunity Services office or "EOS"). Please know there are places on campus where you can make a report in confidence. You can find more information about resources on the UH Title IX/Sexual Misconduct Resources page. Please note that you may also report concerns of discrimination based on your protected class identity to EOS.

Reasonable Academic Adjustments/Auxiliary Aids

The University of Houston is committed to providing an academic environment and educational programs that are accessible for its students. Any student with a disability who is experiencing barriers to learning, assessment or participation is encouraged to contact the Justin Dart, Jr. Student Accessibility Center (Dart Center) to learn more about

academic accommodations and support that may be available to them. Students seeking academic accommodations will need to register with the Dart Center as soon as possible to ensure timely implementation of approved accommodations. Please contact the Dart Center by visiting the website: https://uh.edu/accessibility/ calling (713) 743-5400, or emailing jdcenter@Central.UH.EDU.

The Student Health Center offers a Psychiatry Clinic for enrolled UH students. Call 713-743-5149 during clinic hours, Monday through Friday 8 a.m. - 4:30 p.m. to schedule an appointment.

The A.D. Bruce Religion Center offers spiritual support and a variety of programs centered on well-being.

The Center for Student Advocacy and Community (CSAC) is where you can go if you need help but don't know where to start. CSAC is a "home away from home" and serves as a resource hub to help you get the resources needed to support academic and personal success. Through our Cougar Cupboard, all students can get up to 30 lbs of FREE groceries a week. Additionally, we provide 1:1 appointments to get you connected to on- and off-campus resources related to essential needs, safety and advocacy, and more. The Cougar Closet is a registered student organization advised by our office and offers free clothes to students so that all Coogs can feel good in their fit. We also host a series of cultural and community-based events that foster social connection and helps the cougar community come closer together. Visit the CSAC homepage or follow us on Instagram: @uh_CSAC and @uhcupbrd. YOU belong here.

Cougar Parent and Family Engagement Office

This office, under the Dean of Students Office (https://www.uh.edu/dos/), coordinates resources for Pregnant and Parenting Students (https://www.uh.edu/dos/parenting-students/), Parents and Family members of UH students (https://www.uh.edu/dos/parents/), and Sexual Misconduct Support Services (https://www.uh.edu/dos/sexual-misconduct-support-services/). A description of available support and resources are provided on the respective web pages. The Cougar Parent and Family Engagement Office is located in Student Center South, room B12.

Academic Honesty Policy

High ethical standards are critical to the integrity of any institution, and bear directly on the ultimate value of conferred degrees. All UH community members are expected to contribute to an atmosphere of the highest possible ethical standards. Maintaining such an atmosphere requires that any instances of academic dishonesty be recognized and addressed. The UH Academic Honesty Policy is designed to handle those instances with fairness to all parties involved: the students, the instructors, and the University itself. All students and faculty of the University of Houston are responsible for being familiar with this policy.

Excused Absence Policy

Regular class attendance, participation, and engagement in coursework are important contributors to student success. Absences may be excused as provided in the University of Houston Undergraduate Excused Absence Policy and Graduate Excused Absence Policy for reasons including medical illness of student or close relative, death of a close family member, legal or government proceeding that a student is obligated to attend, recognized professional and educational activities where the student is presenting, and University-sponsored activity or athletic competition. Under these policies, students with excused absences will be provided with an opportunity to make up any quiz, exam or other work that contributes to the course grade or a satisfactory alternative. Please read the full policy for details regarding reasons for excused absences, the approval process, and extended absences. Additional policies address absences related to military service, religious holy days, pregnancy and related conditions, and disability.

Recording of Class

Students may not record all or part of class, livestream all or part of class, or make/distribute screen captures, without advanced written consent of the instructor. If you have or think you may have a disability such that you need to record class-related activities, please contact the Justin Dart, Jr. Student Accessibility Center. If you have an accommodation to record class-related activities, those recordings may not be shared with any other student, whether in this course or not, or with any other person or on any other platform. Classes may be recorded by the instructor. Students may use instructor's recordings for their own studying and notetaking. Instructor's recordings are not authorized to be shared with anyone without the prior written approval of the instructor.

Failure to comply with requirements regarding recordings will result in a disciplinary referral to the Dean of Students Office and may result in disciplinary action.